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Allergic Rhinitis

Definition and Burden

Allergic rhinitis (AR) is a very common entity, affecting 10–30% adults and nearly 40% children at some parts of their life. Almost 80% of asthmatics have associated AR and around 50% of AR has associated asthma.

As per Allergic Rhinitis and its Impact on Asthma (ARIA), AR is defined as:

- ☑ Type 1 hypersensitive inflammation of the nasal mucosa
- ☑ Induced by exposure to allergenic substance
- ☑ With at least two of the four cardinal nasal symptoms, e.g., *sneezing, rhinorrhea, nasal itching,* and *nasal block* present for >1 hour/day on most/many days in a year.

Clinical Classification

Allergic rhinitis is of four types, based on ARIA guidelines as shown in the **Figure 1**. These are mild intermittent, mild persistent, moderate-to-severe intermittent, and moderate-to-severe persistent.

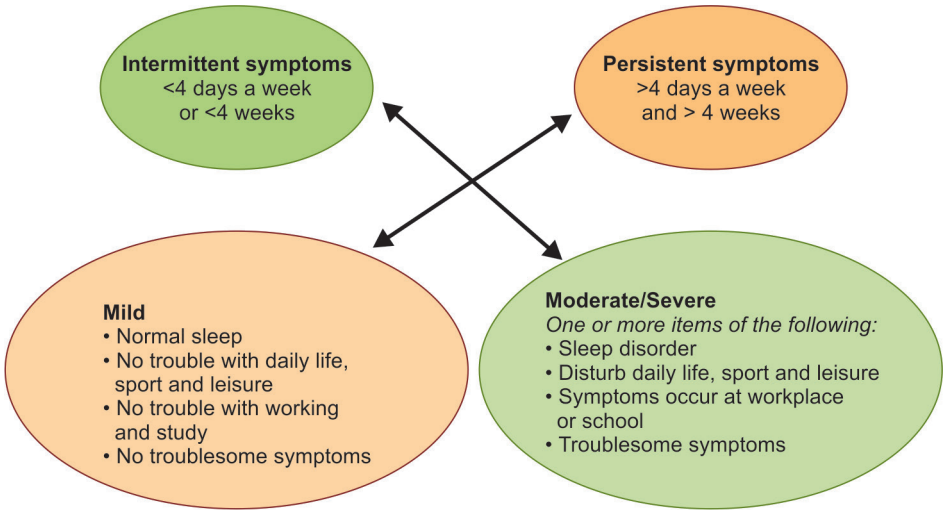


Fig. 1: Allergic Rhinitis and its Impact on Asthma (ARIA) classification according to symptom duration and severity. Persistent allergic rhinitis (AR) should have symptoms for >4 consecutive weeks.

Triggers of Allergic Rhinitis

- ☑ *Allergens*:
  - Aeroallergens-outdoor and indoor allergens
  - Dust, animal, and insect allergens-house dust, house dust mites, cockroach, and pet dander
  - Occupational
  - Latex—important in healthcare professionals
- ☑ *Pollutants*:
  - Indoor—biomass fuel, tobacco, and gas pollutants
  - Outdoor—ozone, oxides of nitrogen, and sulfur from diesel exhaust
- ☑ *Drugs:*
  - Aspirin and other nonsteroidal anti-inflammatory drug (NSAID)

Comorbidities with Allergic Rhinitis

Primary: Asthma, sinusitis, otitis media with effusion, conjunctivitis, laryngitis, atopic dermatitis, recurrent upper respiratory tract infection (URTI), and obstructive sleep apnea (OSA).

Secondary: Decreased quality of life, adenoid hypertrophy, snoring, bruxism, dental malocclusion, learning, and attention impairment.

Diagnosis

- ☑ The diagnosis of Allergic Rhinitis is based on a detailed history, characteristic signs and symptoms and the confirmed presence of allergen specific IgE.
- ☑ Allergen specific IgE testing (either skin prick testing or In vitro specific IgE testing) have a supportive role in the diagnosis. It is very important that these tests should be guided by a careful history and local aerobiology.
- ☑ Tests like total eosinophil count and total IgE lack specificity and sensitivity and is not recommended routinely.

Management

Combination of patient education, self-monitoring, regular physician visits, and avoidance of triggers along with pharmacotherapy is the corner stone in the management **(Flowchart 1)**.

Allergen Avoidance

- ☑ To avoid areas with high pollen, fungal, and dusty environment
- ☑ Rigorous implementation of pest control measures
- ☑ Control of house dust mites (most common, universal allergens)
  - Avoid wall-to-wall carpeting
  - Washing bed sheets in hot water and sun drying, putting mattresses in sun
  - Dust mite encasing materials with very small pore size to prevent mite intrusion
  - Reduction in relative humidity below 50% to reduce house dust mite concentration in the environment

Pharmacological Treatment

- ☑ Second-generation antihistamines are drugs of choice in mild intermittent AR for better safety efficacy ratio **(Table 1)**.
- ☑ The choice of drugs predominantly depends on the dominant symptoms the child suffers from **(Table 2)**.
- ☑ Although intranasal antihistamines are recommended in >6 years in children, but it should better be avoided for its bitter taste and somnolence.
- ☑ Oral decongestants are not recommended in children due to their systemic side effects such as irritability, dizziness, headache, tremor and insomnia, and tachycardia and hypertension.
- ☑ Inhaled nasal steroids (INS) are the controller medications of choice.
- ☑ Choose an INS with low systemic bioavailability like mometasone or fluticasone furoate at a minimum dose required to achieve symptom control **(Table 3)**. Minimum recommended age is >2 years.

TABLE 1: Second-generation antihistamines: Recommended age and dosage.

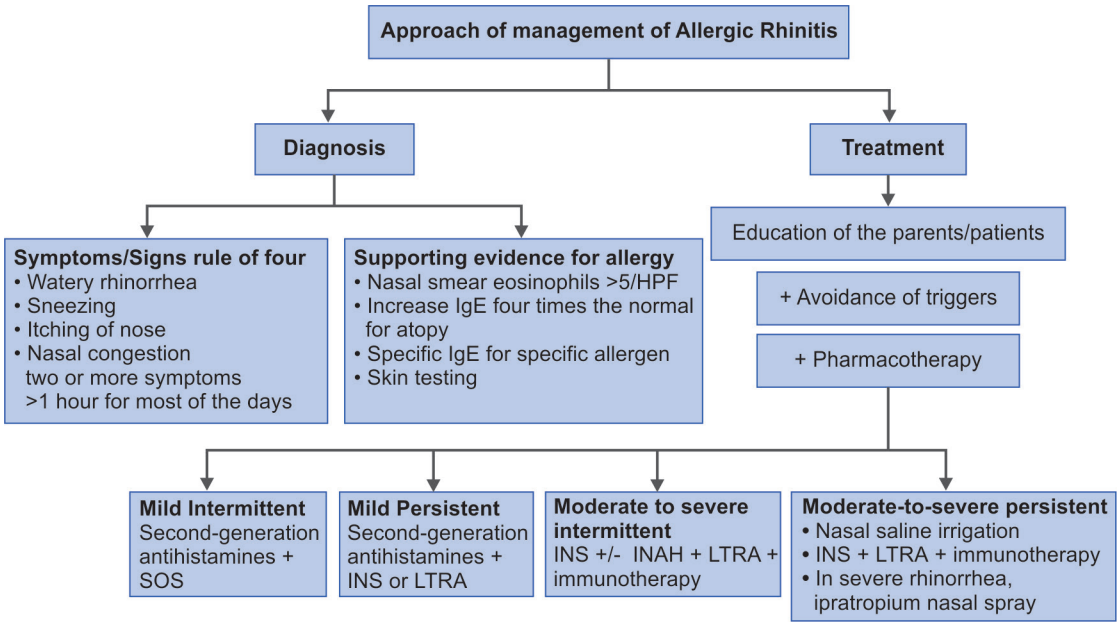
Drug	*Age*	*Dose (mg)*
Cetirizine	>6 months	2.5–10
Desloratadine	>6 months	1.25–5
Fexofenadine	>6 months	30–180
Levocetirizine	>2 years	2.5–10
Loratadine	>2 years	5–10
Ebastine	>2 years	5–20
Rupatadine	>2 years	2.5–10
Bilastine	>6 years	10–20

- ☑ Technique of INS should be evaluated during each visit.
- ☑ Allergen-specific immunotherapy (AIT or desensitization) is recommended in difficult to treat moderate-to-severe and/or persistent AR by a trained allergist.
- ☑ Sublingual immunotherapy (SLIT) and subcutaneous immunotherapy (SCIT) are the two most common modes of immunotherapy.

TABLE 2: Characteristics of drugs used in allergic rhinitis.

Drugs	*Nasal itching and sneezing*	*Rhinorrhea*	*Nasal obstruction*	*Impaired smell*
Sodium cromoglycate	+	+	+/–	–
Topical decongestants	–	–	+++	–
Oral antihistamines	+++	++	+/–	–
Ipratropium bromide	–	+++	++	+
Topical corticosteroids	+++	+++	++	+
Oral corticosteroids	+++	+++	+++	++

Flowchart 1: Practical stepwise approach of management of allergic rhinitis (ARIA, 2019).



(ARIA: Allergic Rhinitis and its Impact on Asthma; HPF: high power field; IgE: immunoglobulin E; INAH: intranasal antihistamine; INS: inhaled nasal steroids; LTRA: leukotriene receptor antagonist)

TABLE 3: Commonly used intranasal corticosteroids (INS) in allergic rhinitis and dosage.

Drugs	*Dose*
Mometasone furoate	2–12 years—50 µg/nostril OD >12 years—100 µg/nostril OD
Fluticasone propionate	4–12 years—50 µg/nostril OD >12 years—100 µg/nostril OD
Fluticasone furoate	4–12 years—27.5 µg/nostril OD >12 years—55 µg/nostril OD

NB: Duration of INS therapy is symptom oriented and highly individualised. May require for 3 months and beyond (does not need weaning).

Monitoring and Follow-up

Regular monitoring of the child is vital for success of therapy. Maintain minimal or low dose of INS to keep patient symptom free.

Conclusion

Allergic rhinitis is a very common respiratory disease affecting the quality of life and often overlooked, but can be managed effectively with a rational approach.

Further Reading

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